

Dr. Stefan Gugler

Curriculum Vitae

Personal Data

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Affiliation Machine Learning Group, TU Berlin / BIFOLD
Nationality Swiss
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Education

- 2023 – **Postdoctoral Studies, TU Berlin/BIFOLD**
Advisor: Klaus-Robert Müller
Since 2024 SNFS Postdoc.Mobility funded
- 2019 – 2023 **Doctoral Studies, ETH Zürich**
Machine learning with physics-based descriptors for quantum chemistry
Advisor: Prof. Markus Reiher
- 2018 **Master Thesis, Massachusetts Institute of Technology (MIT)**
Machine Learning in Enumerative Inorganic Chemistry
Advisor: Prof. Heather Kulik
- 2016 – 2018 **MSc Interdisciplinary Sciences, ETH Zürich**
Major: Computational and Physical Chemistry
- Semester Thesis: *Dispersion Corrections for Density Functional Theory from Gaussian Processes* (Advisor: Prof. Markus Reiher)
 - Semester Thesis: *Jury Evaluation of an Artificial Neural Network Ensemble for Antimicrobial Peptide Prediction* (Advisor: Prof. Gisbert Schneider)
- 2013 – 2016 **BSc Interdisciplinary Sciences, ETH Zürich**
Major: Biochemistry
- Bachelor Thesis: *Towards the Synthesis of a Push-Pull [5,5]Fulvalene-[3]Cumulene* (Advisor: Prof. François Diederich)
 - Semester Thesis: *Construction of Higher Order Protein Cages with O3-33-neg6* (Advisor: Prof. Donald Hilvert)

■ Funding

- 2019 Schweizerische Studienstiftung Study Abroad Grant (MSc Thesis, MIT; 2,000 CHF)
- 2024 Swiss National Science Foundation (SNF), Postdoc.Mobility grant (115,000 CHF)

■ Fellowships

- 2012 Swiss Youth in Science Alumnus
- 2013 Stiftung Dialog "Jugend Debattiert" (Youth Debating Organization) Alumnus
- 2014 Swiss Talent Forum Alumnus
- 2012–2023 Schweizerische Studienstiftung (Swiss Study Foundation, 0.3 % of Swiss students)

■ Awards

- 2011 Science on the Move (biology and biochemistry), 2nd rank
- 2012 Swiss Youth in Science (SJf), predicate "very good"
- 2019 Best Poster Award, 9th Molecular Quantum Mechanics Conference, Heidelberg
- 2020 Outstanding Early Career Paper Award (MSDE) with Prof. Dr. Heather Kulik
- 2025 Dataset Proposal Competition (NeurIPS, AI4Science Workshop)
- 2026 Outstanding Reviewer Awards 2025 for *Machine Learning: Science and Technology* (IOP Publishing)

■ Teaching

- Fall 2019 Introduction to Physical Chemistry, ETH Zürich
- Spring 2019 Quantum Chemistry, ETH Zürich
- Fall 2020 Advanced Quantum Chemistry, ETH Zürich
- Spring 2021 Quantum Chemistry, ETH Zürich
- Fall 2021 Introduction to Physical Chemistry, ETH Zürich
- Spring 2022 Statistical Thermodynamics, ETH Zürich
- Fall 2022 Introduction to Physical Chemistry, ETH Zürich
- Spring 2023 Molecular Quantum Mechanics, ETH Zürich
- Spring 2024 Project Machine Learning, TU Berlin
- 2025–2026 *SNF does not allow teaching*

■ Supervision & Mentoring

- PhD co-supervision
- 2026– David Bechtold, co-supervision; generative ML for spin reaction chemistry
- 2024– Max Eissler, full supervision; ML molecular dynamics and generative chemistry
- 2024–2026 Morris Trestman, full supervision; sampling and active learning for quantum chemistry
- 2023–2025 Khaled Kahouli, shared co-supervision; diffusion models for chemistry and materials
- 2023–2025 Jonas Lederer, shared co-supervision; explainable AI and optimization

MSc / BSc supervision

- 2024 Lukas Hönig (MSc), *Equivariant Neural Networks for Orbital-Free Density Functional Theory*
- 2024 Maxim Hirschmann (BSc), *Enhancing Potential Energy Surface Optimization in Machine Learning through Second Order Derivatives*
- 2025 Simon Bienewald (MSc), *xSchNet: A self-explainable AI-model for quantum chemistry at orbital level*
- 2025 Philipp Pirlet (MSc), *Meta Learning for Wave Function Prediction in Quantum Systems*
- 2026 Sven Elyes Leschber (MSc), *Impact of model expressivity on conformer generation*

Language Skills

- German Native
- English C2 (Certificate in Advanced English (CAE), 2012)
- French B2 (Diplôme d'études en langue française B2 (DELF), 2011)
- Latin B2 (Latinum, 2012)

Volunteer

- 2010 – 2019 Volunteer at Lateintag.ch, for the preservation of Latin
- 2015 – 2023 Board member at reatch.ch, science communication think tank
- 2016 – 2017 Board member & organization of Denkfest Zürich
- 2018 Board Member of VAC, Union of Assistants at the Chemical Labs of ETH Zürich

Invited Talks

- 10/2022 **Exploiting machine learning for quantum chemistry**
QuNB Group Meeting (Stijn De Baerdemacker), New Brunswick (Canada), University of New Brunswick
- 11/2022 **Quantum chemical molecular similarity descriptors for machine learning**
Data-Driven Chemistry Group (Jonny Proppe), Braunschweig (Germany), Technische Universität Braunschweig
- 12/2023 **Quantum chemical roots of molecular descriptors for machine learning**
Computational Materials Design Group (Janine George), Berlin (Germany), Bundesanstalt für Materialforschung und -prüfung
- 11/2024 **Exploring chemical reaction networks with diffusion probabilistic models**
Computational Chemistry Group (Johannes Kästner), Stuttgart (Germany), University of Stuttgart
- 12/2024 **Accelerating computational chemistry with machine learning: From reverse diffusion to explainable AI**
Fakultät für Chemie und Pharmazie (Thorsten Hugel), Freiburg im Breisgau (Germany), University of Freiburg
- 06/2025 **Finding the formose cycle in a diffusion-explored chemical reaction network**
Theoretical Structural Biology Group (Ariane Nunes-Alves), Berlin (Germany), Technical University Berlin
- 08/2025 **Generative computational chemistry: From reverse diffusion to XAI**
Department of Chemistry (Markus Meuwly), Basel (Switzerland), University of Basel
- 09/2025 **Chemical reaction network exploration with probabilistic diffusion models**
DTU Energy (Tejs Vegge), Copenhagen (Denmark), Technical University of Denmark
- 10/2025 **Machine learning and AI for the sciences – toward understanding**
Latvian Institute of Organic Synthesis (Aigars Jirgensons), Riga (Latvia)
- 06/2026 **Generative exploration of chemical reaction networks**
Laboratoire de Physique et Chimie Théoriques (Francesco Talotta), Nancy (France), CNRS / Université de Lorraine

Contributed Talks

- 11/2019 **Gaussian process-based sampling and refinement for dispersion correction**
IBM Research C4 Workshop, Zürich (Switzerland)
- 12/2019 **Gaussian process-based refinement of weak intermolecular forces**
MRS Fall Meeting 2019, Boston (USA)
- 09/2022 **Exploiting machine learning for quantum chemistry**
Colloquium of Laboratory of Physical Chemistry, Zürich (Switzerland), ETH Zürich
- 09/2024 **Finding molecular minima and transition states with reverse diffusion**
Symposium on Theoretical Chemistry, Braunschweig (Germany), TU Braunschweig

- 09/2025 **Chemical reaction network exploration with diffusion probabilistic models**
Young WATOC, Oslo (Norway)
- 12/2025 **How simple can you go? Machine learning molecular dynamics with minimal physical constraints**
Pacifichem 2025, Honolulu (USA)

Posters

- 08/2018 **Enumerating the inorganic universe of small complexes for machine learning**
256th ACS National Meeting & Exposition, Boston (USA)
- 06/2019 **Gaussian Process-Based Refinement of Dispersion Correction**
9th Molecular Quantum Mechanics Conference, Heidelberg (Germany)
- 09/2019 **Gaussian Process-Based Refinement of Dispersion Correction**
SCS Fall Meeting, Zürich (Switzerland)
- 06/2022 **Gaussian Process-Based Refinement of Dispersion Correction**
CECAM: Error control in first-principles modelling, Lausanne (Switzerland)
- 07/2022 **Molecular descriptors rooted in physical principles for ML energies**
12th Triennial Congress of the World Association of Theoretical and Computational Chemists, Vancouver (Canada)
- 09/2022 **Molecular descriptors rooted in physical principles for ML energies**
SCS Fall Meeting, Zürich (Switzerland)
- 03/2025 **Finding Molecular Minima and Transition States with Reverse Diffusion**
CECAM: Density Functional Theory and Artificial Intelligence learning from each other, Lausanne (Switzerland)
- 09/2026 **How simple can you go? A “physics-free” approach to molecular dynamics**
Symposium on Theoretical Chemistry, Berlin (Germany), FU Berlin
- 12/2026 **DMRG quantum chemistry dataset for multi-reference machine learning**
NeurIPS (AI4Science Workshop), San Diego (USA)

Previous and Current Cooperation Partners

- Academic Prof. Dr. Klaus-Robert Müller (TU Berlin / BIFOLD), Prof. Dr. Markus Reiher (ETH Zürich), Prof. Dr. Heather J. Kulik (MIT), Prof. Dr. O. Anatole von Lilienfeld and Dr. Felix A. Faber (University of Toronto / Vector Institute), Prof. Dr. Jonny Proppe (TU Braunschweig), Prof. Dr. Pavlo O. Dral (Xiamen University), Dr. Niklas W. A. Gebauer and Dr. Shinichi Nakajima (BIFOLD)
- Industry Google DeepMind, Berlin (Dr. Oliver T. Unke, Dr. Stefan Ganscha): transformer molecular dynamics, diffusion models, QCML dataset; Microsoft Research AI for Science (Prof. Dr. Frank Noé): generative force fields; BASLEARN, the BASF–TU Berlin joint laboratory for machine learning

Review service

- 2025– AISTATS Conference

Journal of Chemical Information and Modeling
Machine Learning: Science and Technology (Trusted Reviewer status)
Nature Computational Science
JACS Au